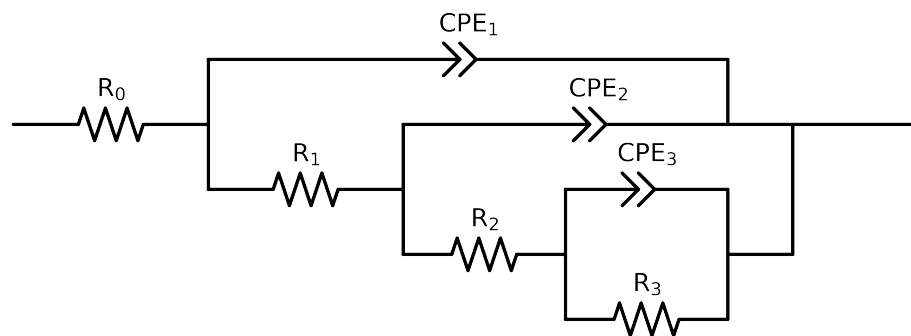


# Comparative EIS Kinetics Report

**Model Structure:** R0-p(R1-p(R2-p(R3,CPE3),CPE2),CPE1)

**Used data:** original data



| Element  | Unit                        | Bounds               | C64_<br>24h_<br>VO3_<br>0.05MCl_<br>4 | C64_<br>24h_<br>VO3_<br>0.05MCl_<br>6 |
|----------|-----------------------------|----------------------|---------------------------------------|---------------------------------------|
| $R_0$    | $\Omega$                    | $[1.0e+00, 1.0e+03]$ | $1.70e+02 \pm 5.2e-01$                | $1.57e+02 \pm 3.6e-01$                |
| $R_1$    | $\Omega$                    | $[1.0e+00, 1.0e+08]$ | $6.45e+01 \pm 1.1e+01$                | $1.57e+02 \pm 2.9e+01$                |
| $R_2$    | $\Omega$                    | $[1.0e+00, 1.0e+08]$ | $1.20e+04 \pm 2.2e+02$                | $2.04e+04 \pm 2.2e+03$                |
| $R_3$    | $\Omega$                    | $[1.0e+00, 1.0e+08]$ | $1.00e+08 \pm 2.6e-06$                | $1.60e+05 \pm 2.1e+04$                |
| $Q_3$    | $\Omega^{-1} \text{ sec}^a$ | $[1.0e-09, 1.0e-03]$ | $2.28e-04 \pm 4.4e-06$                | $9.65e-05 \pm 6.0e-06$                |
| $n_3$    | -                           | $[5.0e-01, 1.0e+00]$ | $5.11e-01 \pm 5.7e-03$                | $5.00e-01 \pm 2.4e-02$                |
| $Q_2$    | $\Omega^{-1} \text{ sec}^a$ | $[1.0e-09, 1.0e-02]$ | $4.35e-05 \pm 7.8e-06$                | $2.86e-05 \pm 7.2e-06$                |
| $n_2$    | -                           | $[5.0e-01, 1.0e+00]$ | $7.07e-01 \pm 1.1e-02$                | $6.91e-01 \pm 2.2e-02$                |
| $Q_1$    | $\Omega^{-1} \text{ sec}^a$ | $[1.0e-09, 1.0e-03]$ | $1.58e-05 \pm 7.3e-06$                | $4.34e-05 \pm 8.9e-06$                |
| $n_1$    | -                           | $[5.0e-01, 1.0e+00]$ | $7.58e-01 \pm 4.3e-02$                | $6.32e-01 \pm 1.8e-02$                |
| $\chi^2$ | -                           | -                    | $1.665e-05$                           | $1.430e-05$                           |

Analysis Results: C64\_24h\_VO3\_0.05MCl\_4

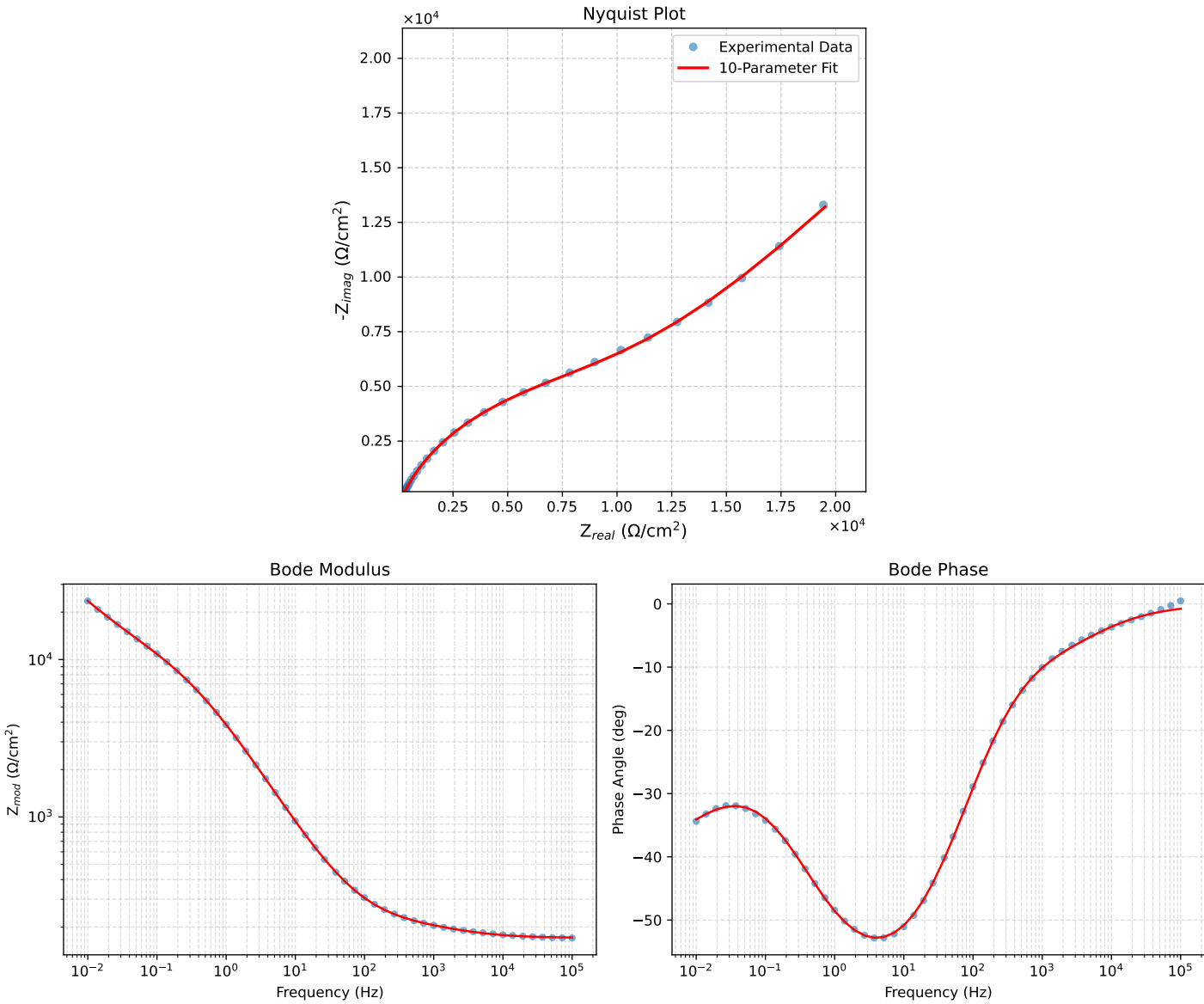


Figure 1: EIS Fit Results for C64\_24h\_VO3\_0.05MCl\_4

# Fitted Data: C64\_24h\_VO3\_0.05MCI\_4

| Frequency (Hz) | $Z_{real}$ ( $\Omega$ ) | $-Z_{imag}$ ( $\Omega$ ) | $Z_{mod}$ ( $\Omega$ ) | $Z_{phase}$ ( $^{\circ}$ ) |
|----------------|-------------------------|--------------------------|------------------------|----------------------------|
| 1.0002e+05     | 1.7141e+02              | 2.3116e+00               | 1.7142e+02             | -0.7727                    |
| 7.2012e+04     | 1.7172e+02              | 2.9378e+00               | 1.7174e+02             | -0.9801                    |
| 5.1855e+04     | 1.7213e+02              | 3.7226e+00               | 1.7217e+02             | -1.2389                    |
| 3.7324e+04     | 1.7267e+02              | 4.7011e+00               | 1.7274e+02             | -1.5595                    |
| 2.6895e+04     | 1.7339e+02              | 5.9035e+00               | 1.7349e+02             | -1.9500                    |
| 1.9395e+04     | 1.7434e+02              | 7.3632e+00               | 1.7449e+02             | -2.4185                    |
| 1.3831e+04     | 1.7563e+02              | 9.1763e+00               | 1.7587e+02             | -2.9909                    |
| 1.0020e+04     | 1.7723e+02              | 1.1210e+01               | 1.7758e+02             | -3.6191                    |
| 7.2070e+03     | 1.7931e+02              | 1.3596e+01               | 1.7983e+02             | -4.3361                    |
| 5.1505e+03     | 1.8196e+02              | 1.6338e+01               | 1.8269e+02             | -5.1309                    |
| 3.7157e+03     | 1.8506e+02              | 1.9295e+01               | 1.8606e+02             | -5.9524                    |
| 2.6867e+03     | 1.8862e+02              | 2.2544e+01               | 1.8997e+02             | -6.8156                    |
| 1.9336e+03     | 1.9268e+02              | 2.6264e+01               | 1.9446e+02             | -7.7620                    |
| 1.3951e+03     | 1.9711e+02              | 3.0585e+01               | 1.9947e+02             | -8.8202                    |
| 1.0007e+03     | 2.0202e+02              | 3.5954e+01               | 2.0519e+02             | -10.0914                   |
| 7.1691e+02     | 2.0745e+02              | 4.2765e+01               | 2.1182e+02             | -11.6479                   |
| 5.2083e+02     | 2.1331e+02              | 5.1119e+01               | 2.1935e+02             | -13.4765                   |
| 3.7287e+02     | 2.2040e+02              | 6.2379e+01               | 2.2905e+02             | -15.8033                   |
| 2.6940e+02     | 2.2859e+02              | 7.6503e+01               | 2.4105e+02             | -18.5041                   |
| 1.9370e+02     | 2.3869e+02              | 9.4892e+01               | 2.5686e+02             | -21.6805                   |
| 1.3923e+02     | 2.5119e+02              | 1.1844e+02               | 2.7771e+02             | -25.2442                   |
| 9.9734e+01     | 2.6703e+02              | 1.4880e+02               | 3.0569e+02             | -29.1274                   |
| 7.2115e+01     | 2.8646e+02              | 1.8620e+02               | 3.4166e+02             | -33.0244                   |
| 5.1511e+01     | 3.1210e+02              | 2.3532e+02               | 3.9088e+02             | -37.0153                   |
| 3.8422e+01     | 3.4032e+02              | 2.8866e+02               | 4.4625e+02             | -40.3052                   |
| 2.6334e+01     | 3.8723e+02              | 3.7543e+02               | 5.3935e+02             | -44.1135                   |
| 1.9290e+01     | 4.3755e+02              | 4.6569e+02               | 6.3900e+02             | -46.7845                   |
| 1.3951e+01     | 5.0465e+02              | 5.8173e+02               | 7.7012e+02             | -49.0584                   |
| 9.9734e+00     | 5.9481e+02              | 7.3047e+02               | 9.4202e+02             | -50.8448                   |
| 7.2338e+00     | 7.0682e+02              | 9.0509e+02               | 1.1484e+03             | -52.0123                   |
| 5.1342e+00     | 8.6280e+02              | 1.1321e+03               | 1.4234e+03             | -52.6877                   |
| 3.7202e+00     | 1.0538e+03              | 1.3885e+03               | 1.7431e+03             | -52.8030                   |
| 2.6893e+00     | 1.3019e+03              | 1.6925e+03               | 2.1353e+03             | -52.4311                   |
| 1.9290e+00     | 1.6287e+03              | 2.0525e+03               | 2.6202e+03             | -51.5669                   |
| 1.3951e+00     | 2.0336e+03              | 2.4482e+03               | 3.1827e+03             | -50.2851                   |
| 9.9777e-01     | 2.5573e+03              | 2.8958e+03               | 3.8633e+03             | -48.5517                   |
| 7.2338e-01     | 3.1704e+03              | 3.3488e+03               | 4.6115e+03             | -46.5669                   |
| 5.1853e-01     | 3.9233e+03              | 3.8255e+03               | 5.4796e+03             | -44.2771                   |
| 3.7321e-01     | 4.7789e+03              | 4.2883e+03               | 6.4209e+03             | -41.9028                   |
| 2.6878e-01     | 5.7294e+03              | 4.7329e+03               | 7.4314e+03             | -39.5589                   |
| 1.9338e-01     | 6.7598e+03              | 5.1632e+03               | 8.5061e+03             | -37.3727                   |
| 1.3918e-01     | 7.8478e+03              | 5.5926e+03               | 9.6366e+03             | -35.4747                   |
| 1.0016e-01     | 8.9834e+03              | 6.0484e+03               | 1.0830e+04             | -33.9519                   |
| 7.1983e-02     | 1.0171e+04              | 6.5677e+03               | 1.2107e+04             | -32.8518                   |
| 5.1807e-02     | 1.1411e+04              | 7.1848e+03               | 1.3485e+04             | -32.1956                   |
| 3.7297e-02     | 1.2730e+04              | 7.9411e+03               | 1.5004e+04             | -31.9565                   |
| 2.6829e-02     | 1.4162e+04              | 8.8801e+03               | 1.6716e+04             | -32.0893                   |
| 1.9312e-02     | 1.5738e+04              | 1.0039e+04               | 1.8668e+04             | -32.5332                   |
| 1.3898e-02     | 1.7507e+04              | 1.1466e+04               | 2.0928e+04             | -33.2215                   |
| 1.0001e-02     | 1.9520e+04              | 1.3209e+04               | 2.3569e+04             | -34.0867                   |

## Analysis Results: C64\_24h\_VO3\_0.05MCl\_6

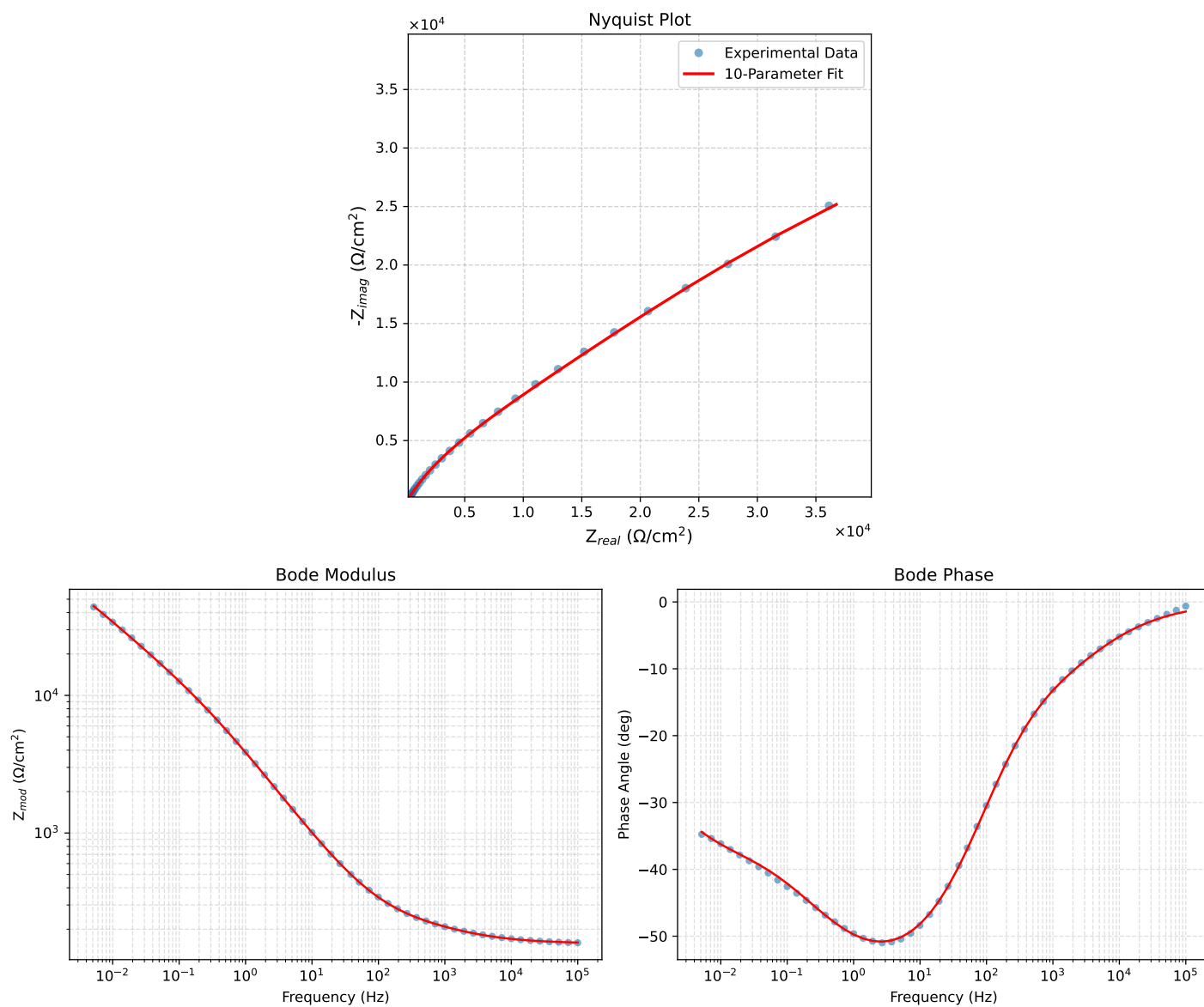


Figure 2: EIS Fit Results for C64\_24h\_VO3\_0.05MCl\_6

# Fitted Data: C64\_24h\_VO3\_0.05MCI\_6

| Frequency (Hz) | $\mathbf{Z}_{real}$ ( $\Omega$ ) | $-\mathbf{Z}_{imag}$ ( $\Omega$ ) | $\mathbf{Z}_{mod}$ ( $\Omega$ ) | $\mathbf{Z}_{phase}$ ( $^\circ$ ) |
|----------------|----------------------------------|-----------------------------------|---------------------------------|-----------------------------------|
| 1.0002e+05     | 1.5999e+02                       | 4.0153e+00                        | 1.6004e+02                      | -1.4377                           |
| 7.2012e+04     | 1.6064e+02                       | 4.9027e+00                        | 1.6071e+02                      | -1.7481                           |
| 5.1855e+04     | 1.6145e+02                       | 5.9743e+00                        | 1.6156e+02                      | -2.1193                           |
| 3.7324e+04     | 1.6245e+02                       | 7.2656e+00                        | 1.6261e+02                      | -2.5609                           |
| 2.6895e+04     | 1.6368e+02                       | 8.8056e+00                        | 1.6392e+02                      | -3.0794                           |
| 1.9395e+04     | 1.6520e+02                       | 1.0632e+01                        | 1.6554e+02                      | -3.6823                           |
| 1.3831e+04     | 1.6714e+02                       | 1.2865e+01                        | 1.6763e+02                      | -4.4017                           |
| 1.0020e+04     | 1.6941e+02                       | 1.5360e+01                        | 1.7010e+02                      | -5.1808                           |
| 7.2070e+03     | 1.7223e+02                       | 1.8314e+01                        | 1.7320e+02                      | -6.0697                           |
| 5.1505e+03     | 1.7571e+02                       | 2.1784e+01                        | 1.7706e+02                      | -7.0670                           |
| 3.7157e+03     | 1.7975e+02                       | 2.5636e+01                        | 1.8157e+02                      | -8.1169                           |
| 2.6867e+03     | 1.8444e+02                       | 2.9986e+01                        | 1.8687e+02                      | -9.2342                           |
| 1.9336e+03     | 1.8994e+02                       | 3.5026e+01                        | 1.9314e+02                      | -10.4483                          |
| 1.3951e+03     | 1.9614e+02                       | 4.0808e+01                        | 2.0034e+02                      | -11.7530                          |
| 1.0007e+03     | 2.0326e+02                       | 4.7754e+01                        | 2.0879e+02                      | -13.2216                          |
| 7.1691e+02     | 2.1128e+02                       | 5.6188e+01                        | 2.1862e+02                      | -14.8927                          |
| 5.2083e+02     | 2.1994e+02                       | 6.6126e+01                        | 2.2966e+02                      | -16.7339                          |
| 3.7287e+02     | 2.3028e+02                       | 7.9102e+01                        | 2.4348e+02                      | -18.9581                          |
| 2.6940e+02     | 2.4197e+02                       | 9.4995e+01                        | 2.5994e+02                      | -21.4348                          |
| 1.9370e+02     | 2.5601e+02                       | 1.1531e+02                        | 2.8078e+02                      | -24.2482                          |
| 1.3923e+02     | 2.7294e+02                       | 1.4093e+02                        | 3.0718e+02                      | -27.3084                          |
| 9.9734e+01     | 2.9387e+02                       | 1.7346e+02                        | 3.4124e+02                      | -30.5515                          |
| 7.2115e+01     | 3.1892e+02                       | 2.1295e+02                        | 3.8348e+02                      | -33.7315                          |
| 5.1511e+01     | 3.5119e+02                       | 2.6399e+02                        | 4.3934e+02                      | -36.9328                          |
| 3.8422e+01     | 3.8584e+02                       | 3.1861e+02                        | 5.0038e+02                      | -39.5483                          |
| 2.6334e+01     | 4.4189e+02                       | 4.0603e+02                        | 6.0011e+02                      | -42.5780                          |
| 1.9290e+01     | 5.0023e+02                       | 4.9556e+02                        | 7.0414e+02                      | -44.7316                          |
| 1.3951e+01     | 5.7571e+02                       | 6.0917e+02                        | 8.3817e+02                      | -46.6180                          |
| 9.9734e+00     | 6.7376e+02                       | 7.5315e+02                        | 1.0105e+03                      | -48.1844                          |
| 7.2338e+00     | 7.9135e+02                       | 9.2080e+02                        | 1.2141e+03                      | -49.3237                          |
| 5.1342e+00     | 9.4913e+02                       | 1.1380e+03                        | 1.4819e+03                      | -50.1710                          |
| 3.7202e+00     | 1.1353e+03                       | 1.3842e+03                        | 1.7902e+03                      | -50.6412                          |
| 2.6893e+00     | 1.3691e+03                       | 1.6797e+03                        | 2.1670e+03                      | -50.8174                          |
| 1.9290e+00     | 1.6678e+03                       | 2.0384e+03                        | 2.6338e+03                      | -50.7112                          |
| 1.3951e+00     | 2.0295e+03                       | 2.4490e+03                        | 3.1807e+03                      | -50.3510                          |
| 9.9777e-01     | 2.4921e+03                       | 2.9426e+03                        | 3.8561e+03                      | -49.7379                          |
| 7.2338e-01     | 3.0358e+03                       | 3.4860e+03                        | 4.6226e+03                      | -48.9489                          |
| 5.1853e-01     | 3.7185e+03                       | 4.1241e+03                        | 5.5529e+03                      | -47.9606                          |
| 3.7321e-01     | 4.5284e+03                       | 4.8316e+03                        | 6.6220e+03                      | -46.8556                          |
| 2.6878e-01     | 5.4863e+03                       | 5.6167e+03                        | 7.8516e+03                      | -45.6729                          |
| 1.9338e-01     | 6.6109e+03                       | 6.4868e+03                        | 9.2619e+03                      | -44.4571                          |
| 1.3918e-01     | 7.9111e+03                       | 7.4450e+03                        | 1.0863e+04                      | -43.2616                          |
| 1.0016e-01     | 9.4025e+03                       | 8.5034e+03                        | 1.2677e+04                      | -42.1257                          |
| 7.1983e-02     | 1.1110e+04                       | 9.6825e+03                        | 1.4737e+04                      | -41.0734                          |
| 5.1807e-02     | 1.3041e+04                       | 1.0992e+04                        | 1.7055e+04                      | -40.1257                          |
| 3.7297e-02     | 1.5233e+04                       | 1.2456e+04                        | 1.9677e+04                      | -39.2731                          |
| 2.6829e-02     | 1.7734e+04                       | 1.4102e+04                        | 2.2657e+04                      | -38.4923                          |
| 1.9312e-02     | 2.0581e+04                       | 1.5938e+04                        | 2.6031e+04                      | -37.7550                          |
| 1.3898e-02     | 2.3841e+04                       | 1.7978e+04                        | 2.9860e+04                      | -37.0195                          |
| 1.0001e-02     | 2.7577e+04                       | 2.0216e+04                        | 3.4194e+04                      | -36.2445                          |
| 7.1974e-03     | 3.1855e+04                       | 2.2630e+04                        | 3.9075e+04                      | -35.3904                          |
| 5.1798e-03     | 3.6742e+04                       | 2.5178e+04                        | 4.4541e+04                      | -34.4217                          |